

Lab Assignment 2

[Start Assignment](#)

- Due Tuesday by 11:59pm
- Points 100
- Submitting a file upload
- Available after Sep 16 at 12am

1. Question: Create an ASCII Sketch Using C++

Write a C++ program that uses cout statements to create an ASCII-art sketch of a **simple house** similar to the one shown below:



Requirements:

- Use only basic C++ output statements.
- Your sketch must contain at least **10 lines**.
- Use spaces and special characters to make the drawing.
- Add your **name** below the sketch.
- The sketch should be neat and properly aligned.

2. Write a C++ program that displays a student ID card containing:

- Student name
- Student ID
- Major
- University
- Year

Use borders made from `*`, `-`, or `=`.

3. Create a restaurant menu using `cout`:

4. Write a C++ program to calculate the total income generated from ticket sales for three seating

categories at a stadium, with the following additional requirements:

- Class A tickets cost \$15 each.
- Class B tickets cost \$10 each.
- Class C tickets cost \$12 each.

Hardcode the number of tickets sold for each class as follows:

- Class A: 50 tickets
- Class B: 30 tickets
- Class C: 40 tickets

The program should:

1. Calculate and display the total income generated from ticket sales.
2. Calculate and display the percentage contribution of each class to the total income.
3. Use the ternary operator to check if the total income is greater than or equal to \$1,000 and display a congratulatory message accordingly.

Grading Breakdown:

1. Hardcoding Ticket Prices (10 Marks):

- 10 Marks for defining constants for ticket prices for Class A, Class B, and Class C.

2. Hardcoding the Number of Tickets Sold (15 Marks):

- 10 Marks for correctly hardcoding the number of tickets sold for Class A, Class B, and Class C.
- 5 Marks for initializing these values correctly in variables.

3. Calculation of Total Income for Each Class (20 Marks):

- 10 Marks for calculating the income for Class A by multiplying the number of tickets by the ticket price.
- 5 Marks for calculating the income for Class B similarly.
- 5 Marks for calculating the income for Class C similarly.

4. Calculation of Overall Total Income (15 Marks):

- 10 Marks for correctly summing up the total income from all three classes.
- 5 Marks for storing the result in a variable and ensuring it is correctly computed.

5. Percentage Contribution Calculation (20 Marks):

- 10 Marks for calculating the percentage contribution of Class A to the total income.
- 5 Marks for calculating the percentage contribution of Class B.
- 5 Marks for calculating the percentage contribution of Class C.

6. Congratulatory Message Display Using Ternary Operator (10 Marks):

- 10 Marks for using the ternary operator to display a congratulatory message if the total income is \$1,000 or more.

7. Congratulatory Message Display Using Ternary Operator (10 Marks):

- **10 Marks for correctly displaying the total income, percentage contributions, and the congratulatory message .**